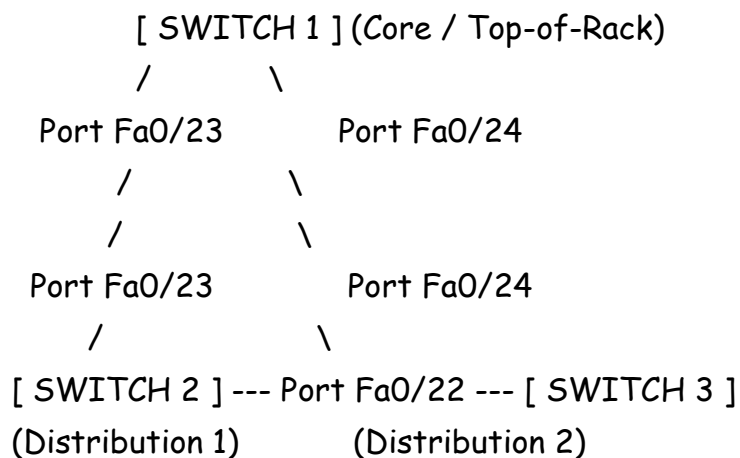


Question: 1

Set up a 3-switch physical network topology using actual switches and Ethernet cables, and draw a labeled diagram of your setup showing all connections?

Answer: 1. Labeled Physical Connections Diagram

text



2. Port-to-Port Wiring Architecture

Link 1 (Switch 1 to Switch 2): Connect a straight-through Ethernet cable from Switch 1 (Port Fa0/23) to Switch 2 (Port Fa0/23).

Link 2 (Switch 1 to Switch 3): Connect a straight-through Ethernet cable from Switch 1 (Port Fa0/24) to Switch 3 (Port Fa0/24).

Link 3 (Switch 2 to Switch 3): Connect a straight-through Ethernet cable from Switch 2 (Port Fa0/22) to Switch 3 (Port Fa0/22).

3. Critical Physical Setup Notes

Cable Type: Modern Gigabit switches use auto-MDIX technology, allowing standard Cat5e or Cat6 Straight-Through Ethernet cables to form your switch-to-switch links flawlessly.

The Spanning Tree Protocol (STP) Behavior: Wiring three switches together in a complete physical loop creates a redundant path. By default, STP will automatically activate inside the switch operating systems to block one of these connections (e.g., the Switch 2 to Switch 3 link) to prevent a catastrophic broadcast storm. If one of the uplink cables to Switch 1 is pulled, the blocked port will safely open within seconds to maintain data flow.

Question: 2

Configure VLANs on each switch so that at least two VLANs exist per switch, and assign at least two ports to each VLAN; take screenshots of your configuration commands or switch console output?

Answer: 1. Create at least two VLANs on each switch, e.g. VLAN 10 and VLAN 20.

2. Assign at least two ports to VLAN 10.

3. Assign at least two ports to VLAN 20.

4. Use these commands:

```
vlan 10
```

```
name VLAN10
```

```
vlan 20
```

```
name VLAN20
```

5. Assign ports using:

```
interface range fa0/1-2
```

```
switchport mode access
```

```
switchport access vlan 10
```

6. For VLAN 20, assign ports similarly using fa0/3-4.

7. Verify the configuration using:

```
show vlan brief
```

show running-config

Question: 3

Enable trunking on the links connecting the switches, and verify trunk status using appropriate show commands; include screenshots of your verification?

Answer: 1. Connect the two switches using the required link/port.

2. Select the port connected to the other switch.

3. Set the port to Trunk mode.

4. Use the command: switchport mode trunk

5. Configure the same trunk settings on both switches.

6. Verify trunk status using: show interfaces trunk

7. Check that the connected port is shown as trunking.

8. Take a screenshot of the show interfaces trunk output as verification.

Question: 4

Create static routes between the VLANs on your network so devices in different VLANs can communicate, and document the routing commands you used?

Answer: 1. Create a static route between the different VLAN networks.

2. Assign IP addresses to the router interfaces for each VLAN.

3. Use the ip route command to configure static routes.

4. Example:

```
ip route 192.168.20.0 255.255.255.0 192.168.1.2
```

```
ip route 192.168.30.0 255.255.255.0 192.168.1.3
```

5. These routes allow devices in different VLANs to communicate.

6. Verify the routing table using:

```
show ip route
```

7. Test communication using ping between devices in different VLANs.

Question: 5

Test end-to-end connectivity by pinging between devices in different VLANs and switches, and include screenshots of successful and failed ping results along with a brief explanation for any failures.

Hint: If pings fail, check VLAN assignments, trunk configuration, and static routes for errors.?

Answer: 1. Test connectivity between devices in different VLANs using the ping command.

2. Ping from one device to another device in a different VLAN.

3. Check whether the ping is successful or failed.

4. Also test connectivity between the switches.

5. A successful ping confirms that VLANs, trunking, and routing are working correctly.

6. If ping fails, check:

VLAN assignments

Trunk configuration

IP addresses and gateways

Static routes

7. Record the successful and failed ping results and include screenshots as proof.

